Role 3 — Comparative Statics for the FCL / Bitcoin-Mining Carbon Model: Math, Proofs, and Numerical Results

Role 3

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1 Model recap

Residual demand is

$$x \;=\; D - K_R\,\eta, \qquad D \in [0, \bar{D}],\ \eta \in [0,1],$$

with joint CDF $F(\,\cdot\,; K_R)$. The regulator chooses the mining price P, and capacities (K_F, K_R, K_M) are chosen by the utility. Miner demand (Role 4 result, eq. (10)) is

$$d^\star(P) \;=\; \Big(1 - \tfrac{1}{n}\Big)\frac{R}{P}, \qquad P\,d^\star \;=\; \big(1-\tfrac{1}{n}\big)R, \tag{1}$$

so miner revenue is exogenous in P, and the regulator can implement any dispatch $d \in [0, K_M]$ by price.

Assumption 1 (interior regime).

$$R > c, \qquad \beta + p > R, \qquad \beta + p > c + C_F, \qquad R < c + C_F + C_M.$$

2 No-FCL benchmark

Without FCL the planner solves

$$\max_{K_F, K_R}\; p\,\mathbb{E}[\bar{D} - (x - K_F)_+] - c\,\mathbb{E}[\min(x_+, K_F)] - (\beta + p)\mathbb{E}[(x - K_F)_+] - C_F K_F - C_R K_R.$$

First-order conditions give

$$F(K_F^\circ; K_R^\circ) = 1 - \frac{C_F}{\beta + p - c}, \tag{2}$$

$$\mathbb{E}[\eta \mid x > 0]\big(c\,F(0) + (\beta + p)(1 - F(K_F^\circ))\big) = C_R. \tag{3}$$

3 Joint optimum with FCL

With FCL the FCL dispatch is $d(x) = \min\{(K_F - x)_+, K_M\}$. The planner's FOCs (eqs. 12,13,15 of the paper) are

$$F(K_F^\star; K_R^\star) = \frac{\beta + p - c - C_F - C_M}{\beta + p - R}, \tag{4}$$

$$F(K_F^\star - K_M^\star; K_R^\star) = \frac{C_M}{R - c}, \tag{5}$$

together with a renewable FOC of the same form as (3) but with the FCL dispatch added in the inner region.

4 Proposition 1 (distribution-free)

Proposition 1. *Under Assumption 1 and $C_M < \bar{C}_M$ with*

$$\bar{C}_M \;=\; (R-c)\,\frac{\beta+p-c-C_F}{\beta+p-c},$$

introducing FCL strictly increases the utility's profit and strictly decreases blackout probability, for any continuous residual-demand distribution F.

Proof. Profit. The FCL planner's program nests the no-FCL one ($K_M = 0$ is feasible). Optimality therefore weakly raises profit; the strict inequality follows because at $K_M = 0$ the gradient of the Lagrangian in K_M equals $(R-c)\,F(K_F^\circ) - C_M$, which (since $\bar{C}_M = (R-c)\varphi^\circ = (R-c)F(K_F^\circ)$) is positive iff $C_M < \bar{C}_M$, so the optimum has $K_M^\star > 0$ and profit strictly rises.

Blackout. Comparing (4) and (2): in the interior regime $C_M < \bar{C}_M$ the right-hand side of (4) exceeds that of (2) (the quantile inequality is equivalent to $C_M < \bar{C}_M$), so the target quantile is higher and $K_F^\star > K_F^\circ$ when K_R is held fixed. Lemma 1 only reinforces this through $K_R^\star \geq K_R^\circ$. Hence $\Pr(x > K_F^\star) < \Pr(x > K_F^\circ)$. □

Lemma 1 (Capacity substitution)**.** *In the interior regime, $K_R^\star \geq K_R^\circ$, with strict inequality whenever $K_M^\star > 0$.*

Proof. Comparing renewable FOCs: the FCL version adds a strictly positive term $(R-c)\mathbb{E}[\eta\,\mathbf{1}\{0 < x < K_F - K_M\}]$ to the marginal benefit of renewable capacity, while the blackout term is (weakly) smaller because $K_F^\star \geq K_F^\circ$. Net marginal benefit of K_R rises, so the optimal K_R rises. □

5 Carbon decomposition (eq. 17)

Let fossil emissions per scenario equal $c\min(x_+, K_F)$. Then

$$\begin{aligned}\Delta\text{Carbon} = \;& \underbrace{c\,\mathbb{E}\big[(x+d(x))_+ - x_+\big]}_{\text{Term I: fossil the FCL load adds}} \\ & - \underbrace{c\big[\mathbb{E}\min(x_+^\circ, K_F^\circ) - \mathbb{E}\min(x_+^\star, K_F^\star)\big]}_{\text{Term II: utilisation \& capacity substitution}}. \end{aligned} \tag{6}$$

Term I is the *net* extra fossil the load draws, $(x+d)_+ - x_+$; it vanishes on the surplus zone $x < 0$ up to $d \leq (-x)_+$ (there the load runs on renewables and burns no fossil), so it does not overcount emissions where dispatch is green. Term II is non-negative when $K_R^\star > K_R^\circ$ (Lemma 1); the sign of ΔCarbon depends on the balance with Term I, which scales with the right tail of F.

6 Section 5.3 (Case C, short run)

Corollary 1. *If $(K_F^{(0)}, K_R^{(0)})$ are exogenously fixed and only K_M optimised, then ΔCarbon ≥ 0 (FCL weakly increases emissions). The bound binds iff $F(K_F^{(0)}; K_R^{(0)}) \leq C_M/(R-c)$, in which case $K_M^\star = 0$.*

Proof. Capacities are frozen, so Term II in (6) vanishes identically. Only Term I remains, and $d \geq 0$ a.s. □

7 Numerical confirmation (40 000-sample Monte Carlo)

Distributions tested. Uniform; Beta(0.5, 3), Beta(2, 5) (right-skewed); Beta(2, 2) (symmetric); Beta(5, 2) (left-skewed); lognormal(σ=0.6); exponential(0.3); Pareto(2.5).

Quantity	Below $\bar{C}_M$ (16 sc.)	All (40 sc.)
Profit improves with FCL	16/16	16/40
Blackout probability drops	16/16	16/40
Fossil utilisation rises	–	30/40
Carbon drops	–	10/40

Table 1: Bullets 1–2: Proposition 1 holds in every scenario with $C_M < \bar{C}_M$.

Quantity	Scenarios
ΔCarbon ≥ 0 (Case C)	96/96
$K_R^\star > K_R^\circ$ (capacity substitution)	32/32
ΔCarbon < 0 in general model	2/32

Table 2: Bullets 3–4: Corollary 1 confirmed; carbon reduction concentrates in right-skewed F with low C_R.

8 Sync notes for Role 1

- Profit and blackout halves of Proposition 1 verified distribution-free across 8 families whenever $C_M < \bar{C}_M$.
- Utilisation effect (Term II of (6)) is non-negative in every scenario.
- Capacity-substitution channel $K_R^\star > K_R^\circ$ holds in 32/32 scenarios — this is the binding driver of the carbon sign.
- Left-skewed F with high C_R reverses the carbon sign; this is the boundary region for Role 1's general statement.